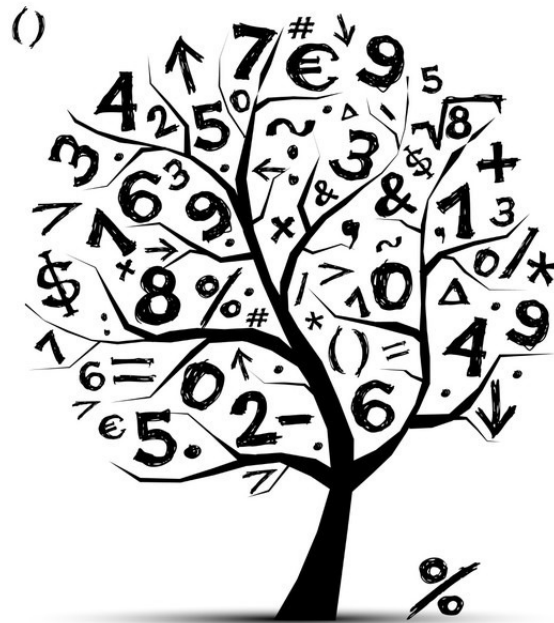


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Exercises: modular arithmetic



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Exercise 1

Find the quotient q and the remainder r (with $0 \leq r < |m|$) of Euclid's division of n by m in the following cases:

- a) $n = 47, m = 5$, b) $n = -47, m = 5$, c) $n = 47, m = -5$, d) $n = -47, m = -5$

In each case we look for the integers q and r such that $n = qm + r$ and $0 \leq r < 5$:

- a) $47 = 9 \cdot 5 + 2$, then $q = 9$ and $r = 2$.
 b) $-47 = (-10) \cdot 5 + 3$, then $q = -10$ and $r = 3$. Note that $-47 = (-9) \cdot 5 - 2$ is not Euclid's division, since $-2 < 0$.
 c) $47 = (-9) \cdot (-5) + 2$, then $q = -9$ and $r = 2$.
 d) $-47 = 10 \cdot (-5) + 3$, then $q = 10$ and $r = 3$.

Exercise 2

Compute:

- a) $100 \bmod 7$, b) $-100 \bmod 7$, c) $2025 \bmod 9$, d) $7 \bmod 12$

By definition, $n \bmod m$ is the remainder of Euclid's division of n by m :

- a) $100 = 14 \cdot 7 + 2$, then $100 \bmod 7 = 2$.
 b) $-100 = (-15) \cdot 7 + 5$, then $-100 \bmod 7 = 5$.
 c) $2025 = 225 \cdot 9 + 0$, then $2025 \bmod 9 = 0$ (2025 is a multiple of 9).
 d) $7 = 0 \cdot 12 + 7$, then $7 \bmod 12 = 7$.

Exercise 3

Establish whether the following congruences hold, using both the definition (same remainder) and the divisibility of the difference:

- a) $38 \equiv 14 \pmod{6}$, b) $17 \equiv 4 \pmod{5}$, c) $100 \equiv 1 \pmod{11}$

- a) $38 = 6 \cdot 6 + 2$ and $14 = 2 \cdot 6 + 2$ have the same remainder 2; equivalently $38 - 14 = 24 = 4 \cdot 6$. The congruence holds.
 b) $17 = 3 \cdot 5 + 2$ and $4 = 0 \cdot 5 + 4$ have different remainders; equivalently $17 - 4 = 13$ is not a multiple of 5. The congruence does not hold.
 c) $100 = 9 \cdot 11 + 1$ and $1 = 0 \cdot 11 + 1$ have the same remainder 1; equivalently $100 - 1 = 99 = 9 \cdot 11$. The congruence holds.

Exercise 4

Let $m \geq 1$ be a natural number. Prove that for every pair of integer numbers n and k :

$$n \bmod m = k \bmod m \iff \text{there exists an integer } q \text{ such that } n - k = q \cdot m$$

($\Rightarrow$) Let $r = n \bmod m = k \bmod m$. By Euclid's division there exist integers q_1 and q_2 such that $n = q_1 m + r$ and $k = q_2 m + r$. Subtracting:

$$n - k = (q_1 - q_2) m$$

so the claim holds with $q = q_1 - q_2$.

($\Leftarrow$) Let $n - k = q m$ and let $k = q_2 m + r$ with $0 \leq r < m$ (Euclid's division of k by m). Then:

$$n = k + q m = (q + q_2) m + r \quad \text{with} \quad 0 \leq r < m$$

By the uniqueness in Euclid's division, r is the remainder of the division of n by m , that is $n \bmod m = r = k \bmod m$.

Exercise 5

Let $m \geq 1$, n, n', k, k' be natural numbers such that $n \equiv n' \pmod{m}$ and $k \equiv k' \pmod{m}$. Prove that:

$$\text{a) } n + k \equiv n' + k' \pmod{m}, \quad \text{b) } n k \equiv n' k' \pmod{m}$$

Then compute $(123 \cdot 456 + 789) \bmod 10$ without computing the product.

By Exercise 4 there exist integers a and b such that $n = n' + a m$ and $k = k' + b m$.

a) $n + k = n' + k' + (a + b) m$, then $(n + k) - (n' + k')$ is a multiple of m .

b) $n k = (n' + a m)(k' + b m) = n' k' + (n' b + k' a + a b m) m$, then $n k - n' k'$ is a multiple of m .

Since $123 \equiv 3$, $456 \equiv 6$ and $789 \equiv 9 \pmod{10}$, we have:

$$123 \cdot 456 + 789 \equiv 3 \cdot 6 + 9 = 27 \equiv 7 \pmod{10}$$

then $(123 \cdot 456 + 789) \bmod 10 = 7$. Check: $123 \cdot 456 + 789 = 56088 + 789 = 56877$.

Exercise 6

Compute:

$$\text{a) } 2^{10} \bmod 7, \quad \text{b) } 3^{100} \bmod 4, \quad \text{c) } 5^{21} \bmod 6$$

We use Exercise 5 b) repeatedly: if $n \equiv n' \pmod{m}$, then $n^p \equiv (n')^p \pmod{m}$ for every natural number $p \geq 1$.

a) $2^3 = 8 \equiv 1 \pmod{7}$, then $2^{10} = (2^3)^3 \cdot 2 \equiv 1^3 \cdot 2 = 2 \pmod{7}$. Then $2^{10} \bmod 7 = 2$ (check: $1024 = 146 \cdot 7 + 2$).

b) $3^2 = 9 \equiv 1 \pmod{4}$, then $3^{100} = (3^2)^{50} \equiv 1^{50} = 1 \pmod{4}$. Then $3^{100} \bmod 4 = 1$.

c) $5^2 = 25 \equiv 1 \pmod{6}$, then $5^{21} = (5^2)^{10} \cdot 5 \equiv 1^{10} \cdot 5 = 5 \pmod{6}$. Then $5^{21} \bmod 6 = 5$.

Exercise 7

Prove that for every integer number n we have $n^2 \bmod 4 \in \{0, 1\}$. Deduce that 2023 is not the square of an integer number.

By Euclid's division of n by 2, every integer n is either even, $n = 2t$, or odd, $n = 2t + 1$, with t integer:

- if $n = 2t$, then $n^2 = 4t^2 = 4t^2 + 0$, so $n^2 \bmod 4 = 0$;
- if $n = 2t + 1$, then $n^2 = 4t^2 + 4t + 1 = 4(t^2 + t) + 1$, so $n^2 \bmod 4 = 1$.

Since $2023 = 505 \cdot 4 + 3$, we have $2023 \bmod 4 = 3 \notin \{0, 1\}$, then 2023 is not the square of an integer number.

Exercise 8

Prove that every natural number is congruent modulo 9 to the sum of its decimal digits. Use this result to compute $123456789 \bmod 9$ and $20251007 \bmod 9$.

A natural number n with decimal digits $c_p, c_{p-1}, \dots, c_1, c_0$ can be written as:

$$n = \sum_{j=0}^p c_j 10^j$$

Since $10 = 1 \cdot 9 + 1$, we have $10 \equiv 1 \pmod{9}$ and then (Exercise 6) $10^j \equiv 1 \pmod{9}$ for every $j \geq 1$ (and trivially for $j = 0$). Using Exercise 5 for each term of the sum:

$$n = \sum_{j=0}^p c_j 10^j \equiv \sum_{j=0}^p c_j \pmod{9}$$

- $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45 = 5 \cdot 9$, then $123456789 \bmod 9 = 0$.
- $2 + 0 + 2 + 5 + 1 + 0 + 0 + 7 = 17 = 1 \cdot 9 + 8$, then $20251007 \bmod 9 = 8$.